**1** Given that $f(x) = x - 4$ find:

(a) f(5) **(1)**

(b) f(3) **(1)**

**(Total for Question 1 is 2 marks)**

**2** Given that $g(x) = 2x^2 - 10$ find:

(a) g(2) **(1)**

(b) g(-2) **(1)**

(c) Solve: $g(x) = 8$ **(3)**

**(Total for Question 2 is 5 marks)**

**3** Given that $f(x) = 3x - 5$ find:

(a) f(3) **(1)**

(b) f(-2) **(1)**

(c) Solve $f(x) = 1$ **(2)**

**(Total for Question 3 is 4 marks)**

**4** Given that $f(x) = x^2 - 3$ find:

(a) f(10) **(1)**

(b) f(-1) **(1)**

(c) Solve: $f^{-1}(x) = 8$ **(2)**

**(Total for Question 4 is 4 marks)**

**5** Given that $f(x) = 2x - 4$ and $g(x) = 3x + 5$

(a) Find gf(3) **(2)**

(b) Work out an expression for $f^{-1}(x)$ **(2)**

(c) Solve $f(x) = g(x)$ **(2)**

**(Total for Question 5 is 6 marks)**

**6** Given that $f(x) = 3x + 1$ and $g(x) = x^2$

(a) Find $fg(x)$ **(2)**

(b) Work out an expression for $gf(x)$ **(2)**

(c) Solve $fg(x) = gf(x)$ **(3)**

**(Total for Question 6 is 7 marks)**

**7** Given that $f(x) = x^2 - 17$ and $g(x) = x + 3$

(a) Work out an expression for $g^{-1}(x)$ **(2)**

(b) Work out an expression for $f^{-1}(x)$ **(2)**

(c) Solve $f^{-1}(x) = g^{-1}(x)$ **(4)**

**(Total for Question 7 is 8 marks)**

**8** The function f is defined such that $f(x) = x^2 - 1$

(a) Find an expression for $f(x - 2)$ **(2)**

(b) Hence solve: $f(x - 2) = 0$ **(2)**

**(Total for Question 8 is 4 marks)**

**9** The function f is defined such that $f(x) = 4x - 1$

(a) Find $f^{-1}(x)$ **(2)**

The function g is defined such that $g(x) = kx^2$ where $k$ is a constant

(b) Given that $fg(2) = 12$
Work out the value of $k$. **(2)**

**(Total for Question 9 is 4 marks)**